

The Interpolated Stochastic Heat and Wave Equation: Solvability and Exact Moment Asymptotics

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Stochastic Calculus

When studying Probability Theory, we consider measurable functions (random variables),

$$X : \Omega \rightarrow \mathbb{R},$$

and we consider the measure spaces

$$(\Omega, \mathcal{F}, \mathbb{P}) \quad \text{and} \quad (\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda),$$

where Ω and $\mathcal{F}$ are referred to as the sample space and set of possible events and

$$\mathbb{P}(\Omega) = 1.$$

The average or expected value of X is

$$\mathbb{E}(X) = \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega) = \int_{\mathbb{R}} x \, dF(x) = \int_{\mathbb{R}} xf(x) \, dx.$$

Stochastic Processes

Definition (Stochastic Process)

A stochastic process is a random variable that is indexed by set, T , which usually represents time:

$$X_t(\omega), \quad t \in T, \omega \in \Omega.$$

One can think a stochastic processes as a two parameter function

$$X_t(\omega) = X(t, \omega) : T \times \Omega \rightarrow \mathbb{R}.$$

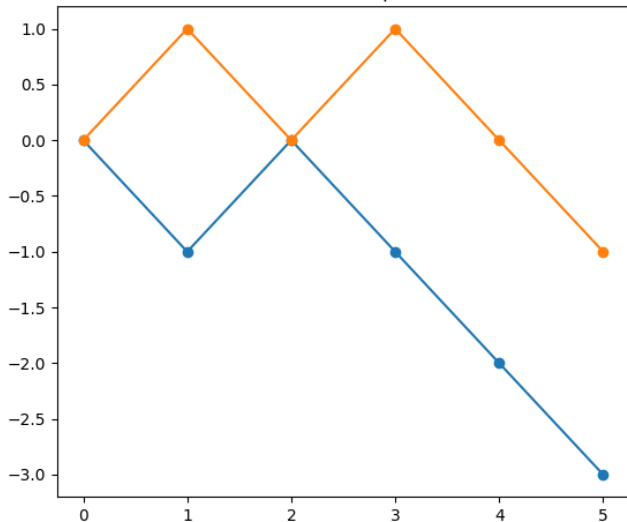
Definition (Sample Path)

The variable ω represents the idea of a single possible outcome. The following function is then referred to as a *sample path*:

$$t \mapsto X_t(\omega).$$

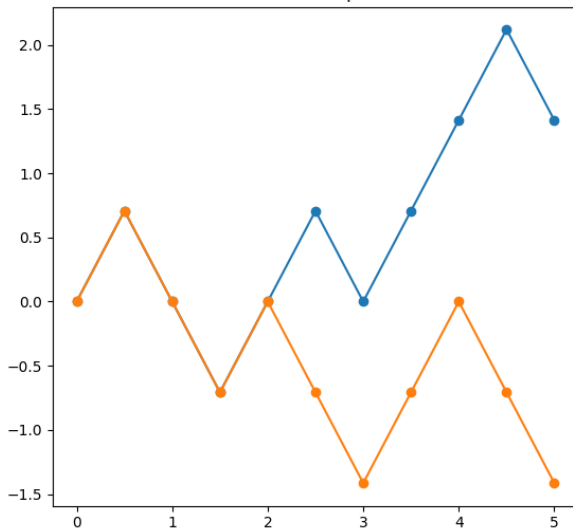
Random Walk Example

2 sample paths of a random walk
jump size = 1.0 and step size = 1.0
number of steps = 5



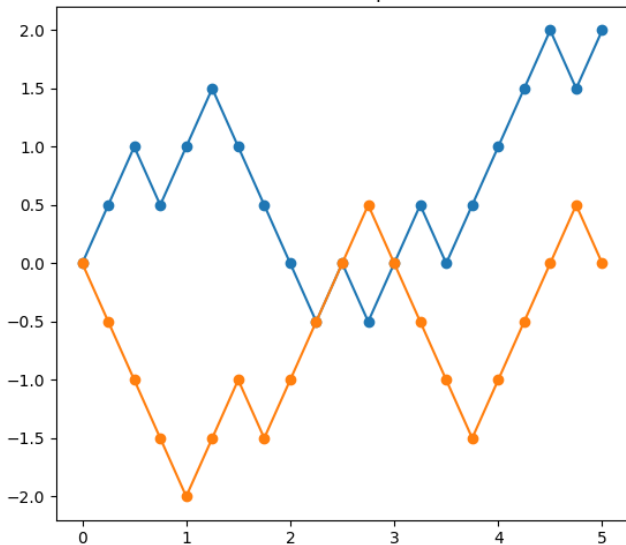
Random Walk Example

2 sample paths of a random walk
jump size = 0.707 and step size = 0.5
number of steps = 10



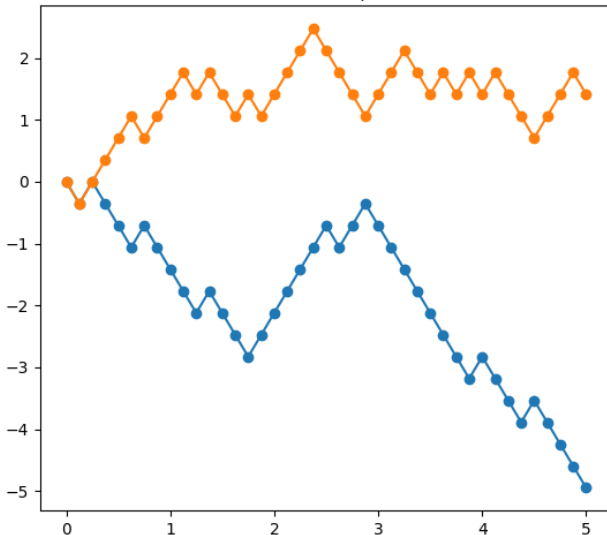
Random Walk Example

2 sample paths of a random walk
jump size = 0.5 and step size = 0.25
number of steps = 20

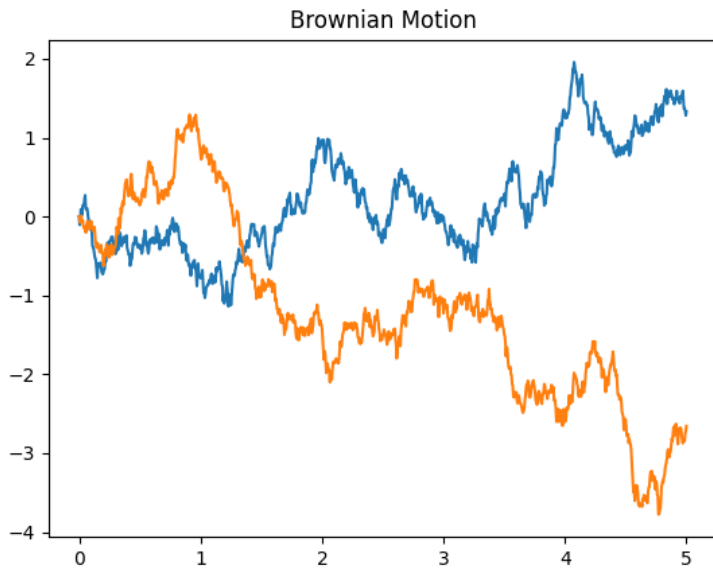


Random Walk Example

2 sample paths of a random walk
jump size = 0.354 and step size = 0.12
number of steps = 40



Random Walk Example



What can we do with Brownian Motion?

Consider a stock that has value S_t at time t . Consider $t = 0$ to be the point of entry into the investment. We can define a return rate as follows:

$$S_t = S_0(1 + R_t), \quad R_t := \frac{S_t - S_0}{S_0} \text{ is the return rate over the period } [0, t].$$

We want to make predictions of our return. In a very simple model, we can assume that

$$R_t \sim \text{Normal}(\mu_t, \sigma^2), \quad \mu \in \mathbb{R}, \sigma \in (0, \infty).$$

In terms of Stochastic Calculus, we express this as the following stochastic differential equation:

$$R_t = \frac{dS_t}{S_t} = \mu dt + \sigma dB_t, \quad \text{where } B_t \text{ is a Brownian motion.}$$

Modeling a stock price with geometric Brownian motion

The notation

$$dS_t = \mu S_t dt + \sigma S_t dB_t, \quad \text{where } B_t \text{ is a Brownian motion}$$

is only symbolic as dB_t does not exist.

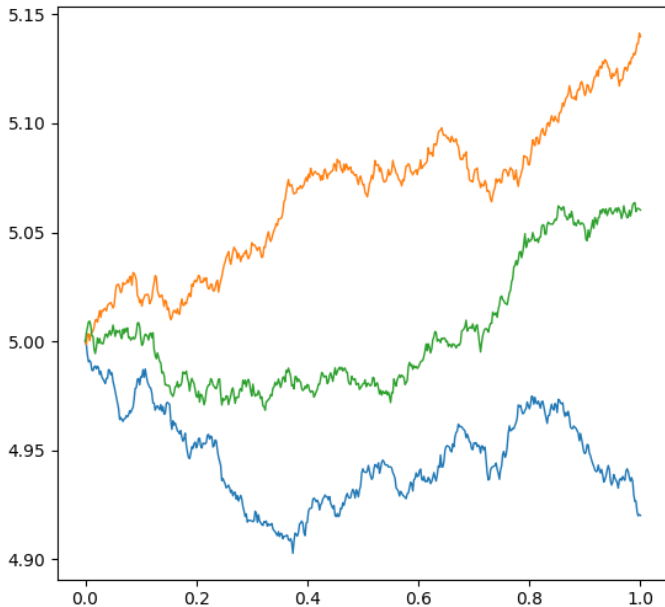
The above can be legitimately viewed as the following integral equation:

$$S_t - S_0 = \int_0^t \mu S_k dk + \int_0^t \sigma S_k dB_k.$$

The stochastic processes S_t that solves the above equation is called a *Geometric Brownian Motion*. Notice that we have come up with a random variable that models the stock price:

$$S_t(\omega) = S_0 + \int_0^t \mu S_k(\omega) dk + \int_0^t \sigma S_k(\omega) dB_k(\omega).$$

Geometric Brownian Motion: $dX_t = \sigma X_t dW_t + \mu X_t dt$
 $(a, b) = (0, 1)$, $X_a = 5$, $\sigma = 0.01$, and $\mu = 0.001$
we subdivide the domain $n = 500$ times



The Itô's integral (a special case – *Wiener integral*)

Suppose that f is a nice continuous function. Then we can define:

$$\int_0^t f(s)dB_s(\omega) := \lim_{\|P\| \rightarrow 0} \sum_{i \in P} f(t_{i-1})(B_{t_i}(\omega) - B_{t_{i-1}}(\omega)),$$

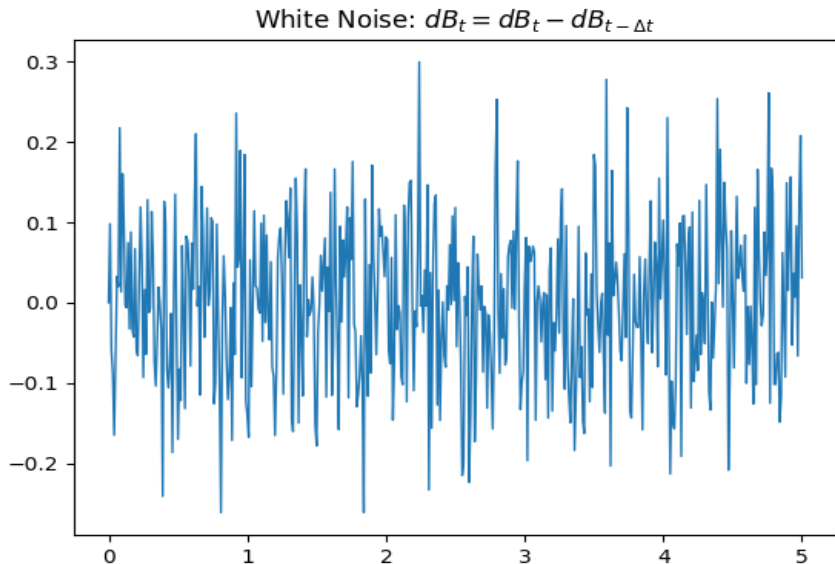
where $P = t_0 = 0, t_1, t_2, \dots, t_n = t$ is a partition of $[0, t]$. In other words, the Itô's integral in this case the left point Riemann sum.

In general, this formula is not true because the increments

$$dB_{t_i}(\omega) := B_{t_i}(\omega) - B_{t_{i-1}}(\omega)$$

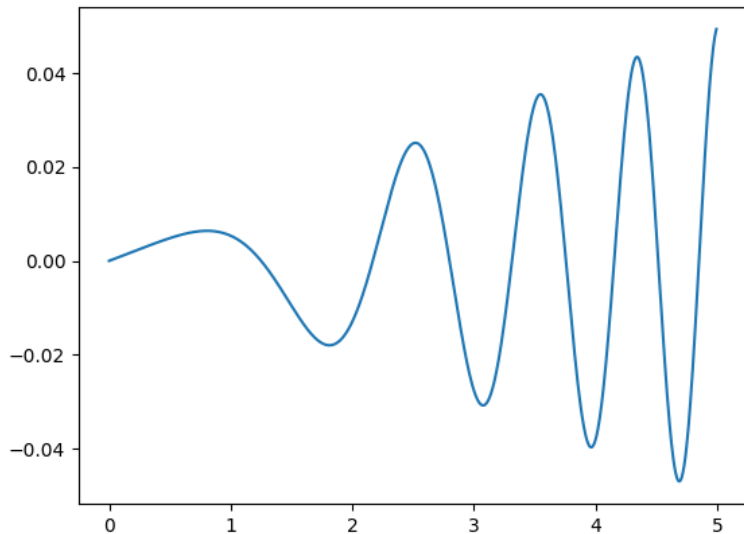
are very bad.

White Noise



Increments of a smooth function for reference

Increments, $df(x_i) = f(x_i) - f(x_{i-1})$, for $f(x) = \sin(x^2)$



A “Fundamental Theorem of Itô’s Calculus”

If f is twice continuously differentiable then we have the following “Fundamental Theorem of Itô’s Calculus”

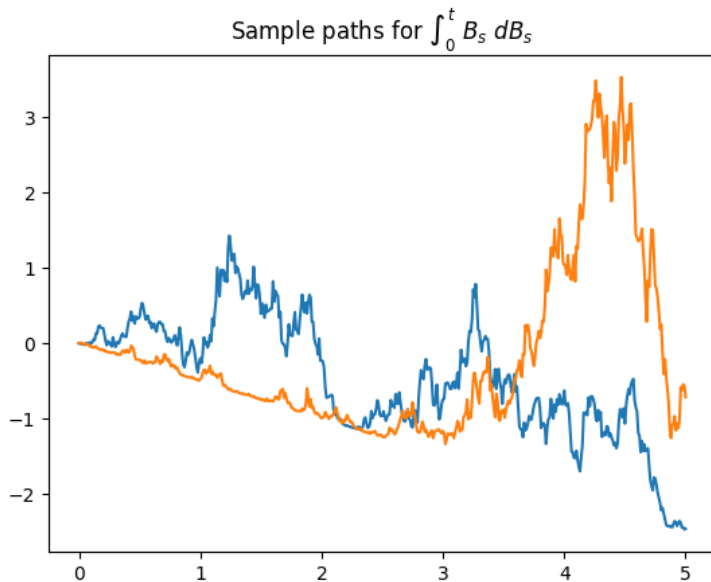
$$\int_a^t \frac{d}{ds} f(B_s) dB_s = f(B_t) - f(B_a) - \frac{1}{2} \int_a^t \frac{d^2}{ds^2} f(B_s) ds.$$

Example

If we apply the above to $f(x) = x^2/2$ with $a = 0$ then we have

$$\int_0^t B_s dB_s = \frac{B_t^2}{2} - \frac{t}{2}.$$

Two sample paths



One more example

For a nice function $f(t, x)$, we can generalize Itô's formula to

$$f(t, B_t) - f(a, B_a) = \int_a^t \frac{\partial f}{\partial x}(s, B_s) dB_s + \int_a^t \left(\frac{\partial f}{\partial t}(s, B_s) + \frac{\partial^2 f}{\partial x^2}(s, B_s) \right) ds.$$

Example

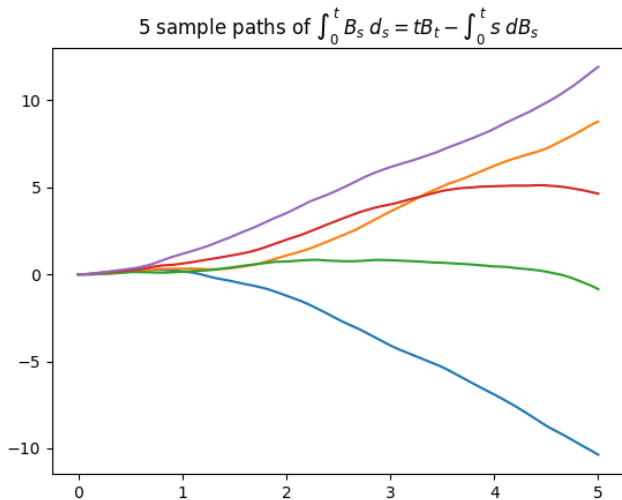
Now consider the function $f(t, x) = tx$. Then

$$\frac{\partial f}{\partial t}(t, x) = x, \quad \frac{\partial f}{\partial x}(t, x) = t, \quad \frac{\partial^2 f}{\partial x^2}(t, x) = 0$$

Plugging this in above for $a = 0$ gives us

$$tB_t = \int_0^t s \, dB_s + \int_0^t B_s \, ds, \quad \implies \quad \int_0^t B_s \, ds = tB_t - \int_0^t s \, dB_s.$$

Smoothness?!



Stochastic Heat Equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \frac{\partial^2}{\partial x^2} u(t, x) = \lambda u(t, x) \frac{\partial^2}{\partial t \partial x} W(t, x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R} \end{cases}$$

The solution to the stochastic heat equation is given through the following integral equation:

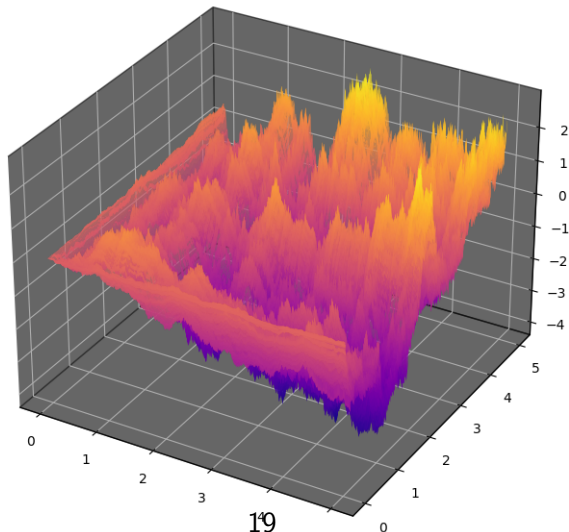
$$u(t, x) = \int_{\mathbb{R}} G(t, x - y) f(y) dy + \lambda \int_0^t \int_{\mathbb{R}} G(t - s, x - y) u(s, y) W(ds, dy),$$

where

$$G(t, x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{|x|^2}{2t}\right)$$

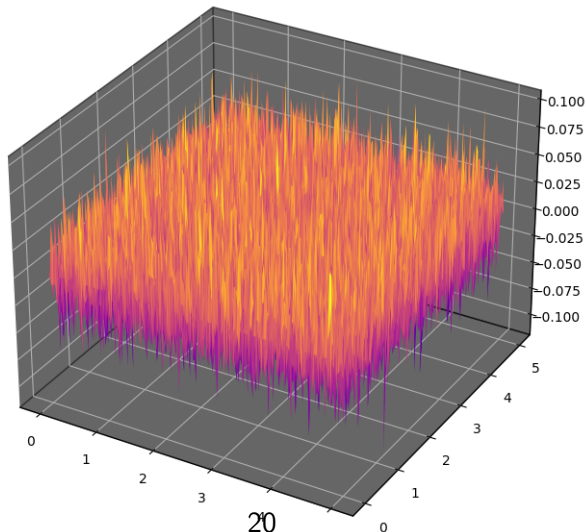
Sample path of a Brownian Sheet, $W(t, x)$

Brownian Sheet



Approximation of a space-time white noise

Space-time white noise: $\dot{W}(t, x) \approx \int_{t-\delta t}^t \int_{x-\delta x}^x dW(t, x)$



Random vs Deterministic

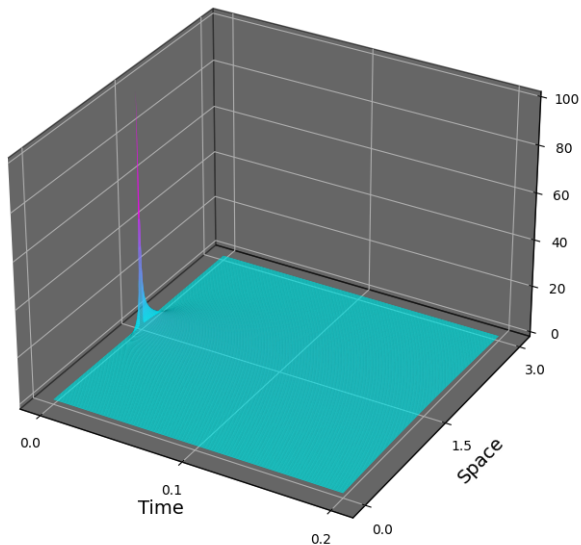
- ▶ The solution to the stochastic heat equation is a stochastic process.

$$u(t, x)(\omega_1), \quad u(t, x)(\omega_2), \quad \cdots, \quad u(t, x)(\omega_n).$$

- ▶ The random solution only exists implicitly through the integral equation.
- ▶ $\frac{\partial}{\partial x} u(t, x)$ and $\frac{\partial}{\partial t} u(t, x)$ does not exist in the random case with space-time white noise.
- ▶ Interesting 'blow up' happens in the random case. This actually has real-world application.

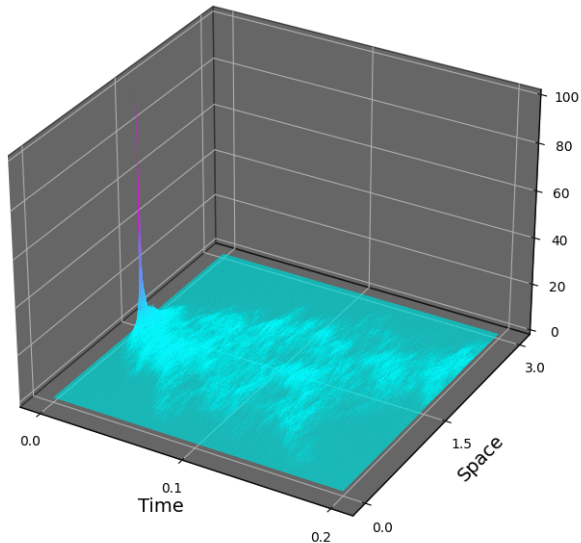
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 0.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



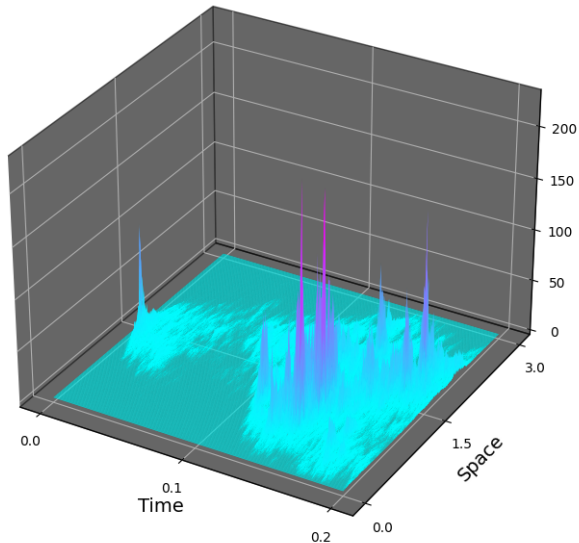
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 1.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



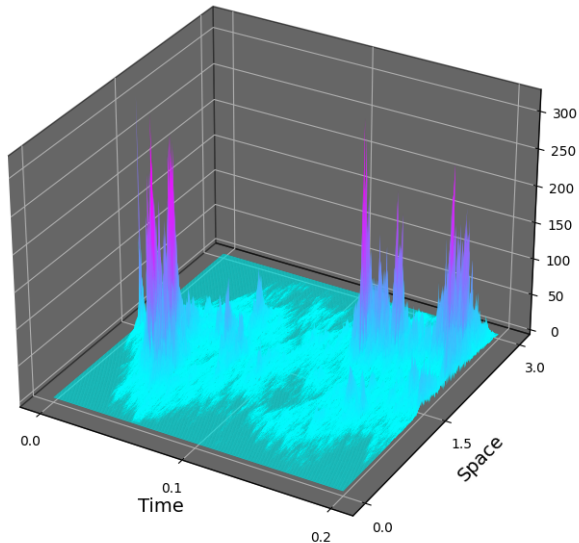
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



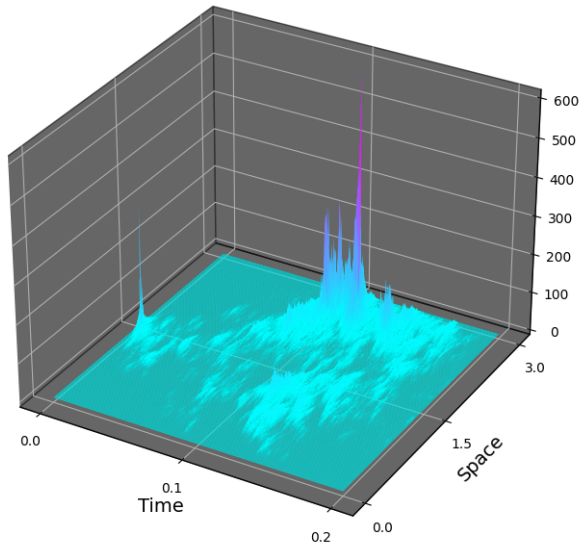
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.05u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



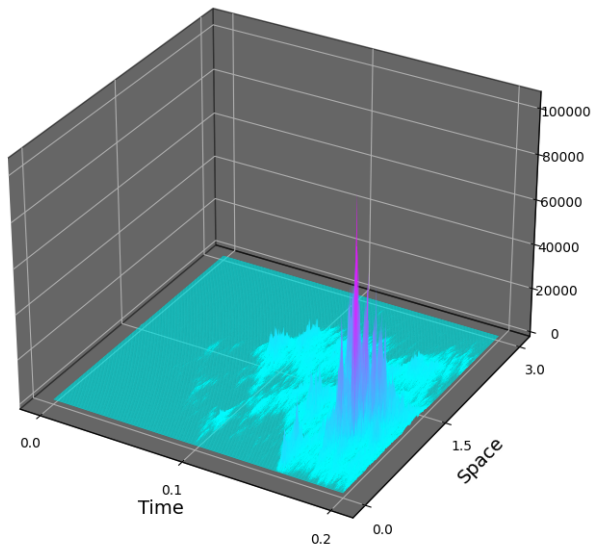
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.15u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.2u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



Growth of Moments

We have a quantity that describes the growth of the p^{th} moments of the solutions to the stochastic heat question:

$$\lim_{t \rightarrow \infty} \frac{\log [\mathbb{E}(|u(t, x)|^p)]}{t}.$$

What does this limit say about the growth or decay of the moments?

Growth of Moments for the Stochastic Heat Equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \frac{1}{2} \frac{\partial^2}{\partial x^2} u(t, x) = \sqrt{\lambda} u(t, x) \frac{\partial}{\partial x} W(x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = 1 & x \in \mathbb{R} \end{cases}$$

Theorem (Hu 2002)

For the above equation, we can prove that

$$\lim_{t \rightarrow \infty} \frac{\log [\mathbb{E}(|u(t, x)|^p)]}{t^3} = \frac{p(p-1)\lambda^2}{24}$$

which means that

$$t \mapsto \mathbb{E}(|u(t, x)|^p) \asymp \exp\left(\frac{p(p-1)\lambda^2}{24} t^3\right), \quad \text{as } t \rightarrow \infty.$$

So the moments grow exponentially in time.